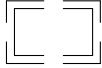
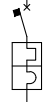
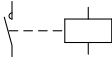
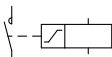
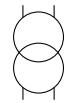
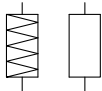


1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																				
SAFETY CRITICAL ELEMENT LIST																				
IFI	1	10/12/2025	AS-BUILT				ADF	GDP	IC											
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC											
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.											
Rev. Index		Date	Description				Prepared	Checked	Approved											
Client							Project Title													
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT													
Contractor							Provision of a m-LNG plant for production for ManaEnergy													
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																				
Engineering																				
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																				
Facility Name			Location				Scale		Sheet of Sheets											
TENDRARA			MOROCCO				A4		1 of 15											
Document Title					Doc. Code															
P-PN-OS371.1					ELD01-ING-FR01															
NGL & CONDENSATE VAPORIZER UNIT					Vendor Doc. code															
					IE24090_SE25_P-PN-OS371.1															
				Project : P–PN–OS371.1			Customer : ManaEnergy Managem Group				Object :									
General Contractor :		Manufacture : esa esaelettromeccanica.com		NGL & CONDENSATE VAPORIZER UNIT			Contract : IE24090				Rev.		Data :		By :					
							S/N : IE24090_SE25				Rev. 1		Data : 10/12/2025		Date : 26/03/2024					
													Sheet : 1		of : 15					

CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

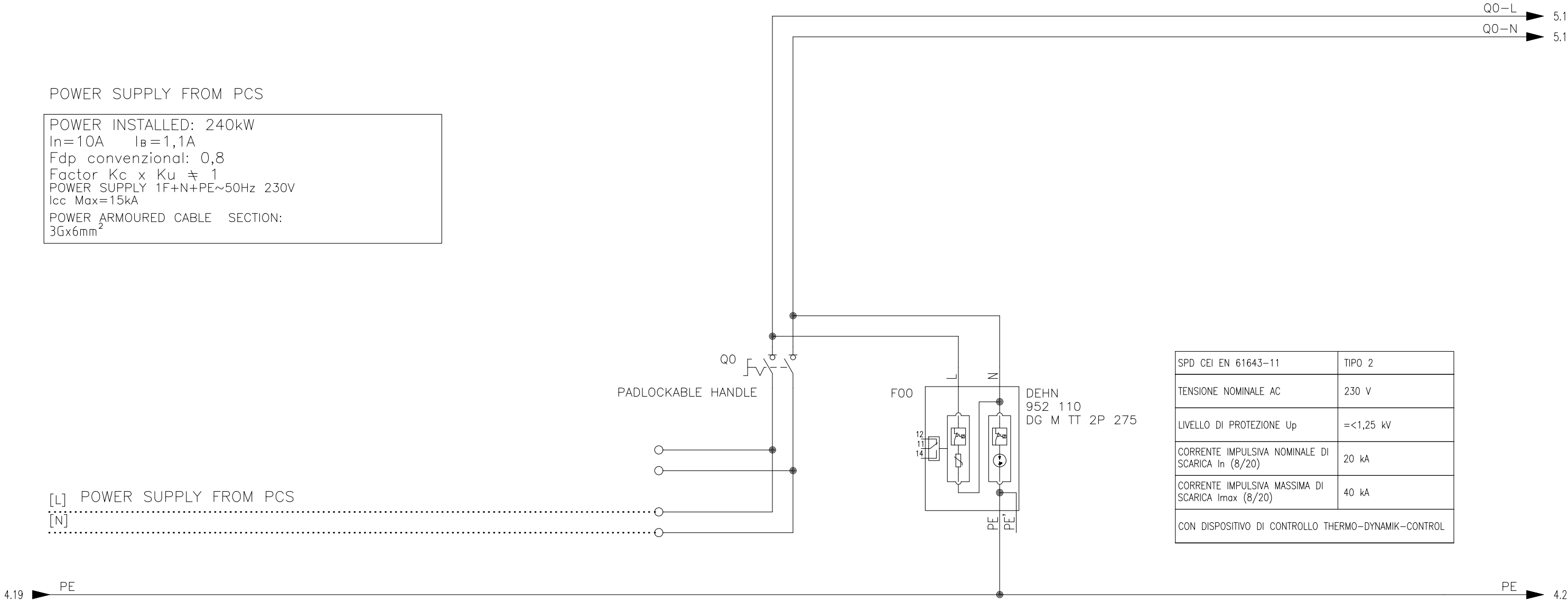
CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT–BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW–GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK–BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

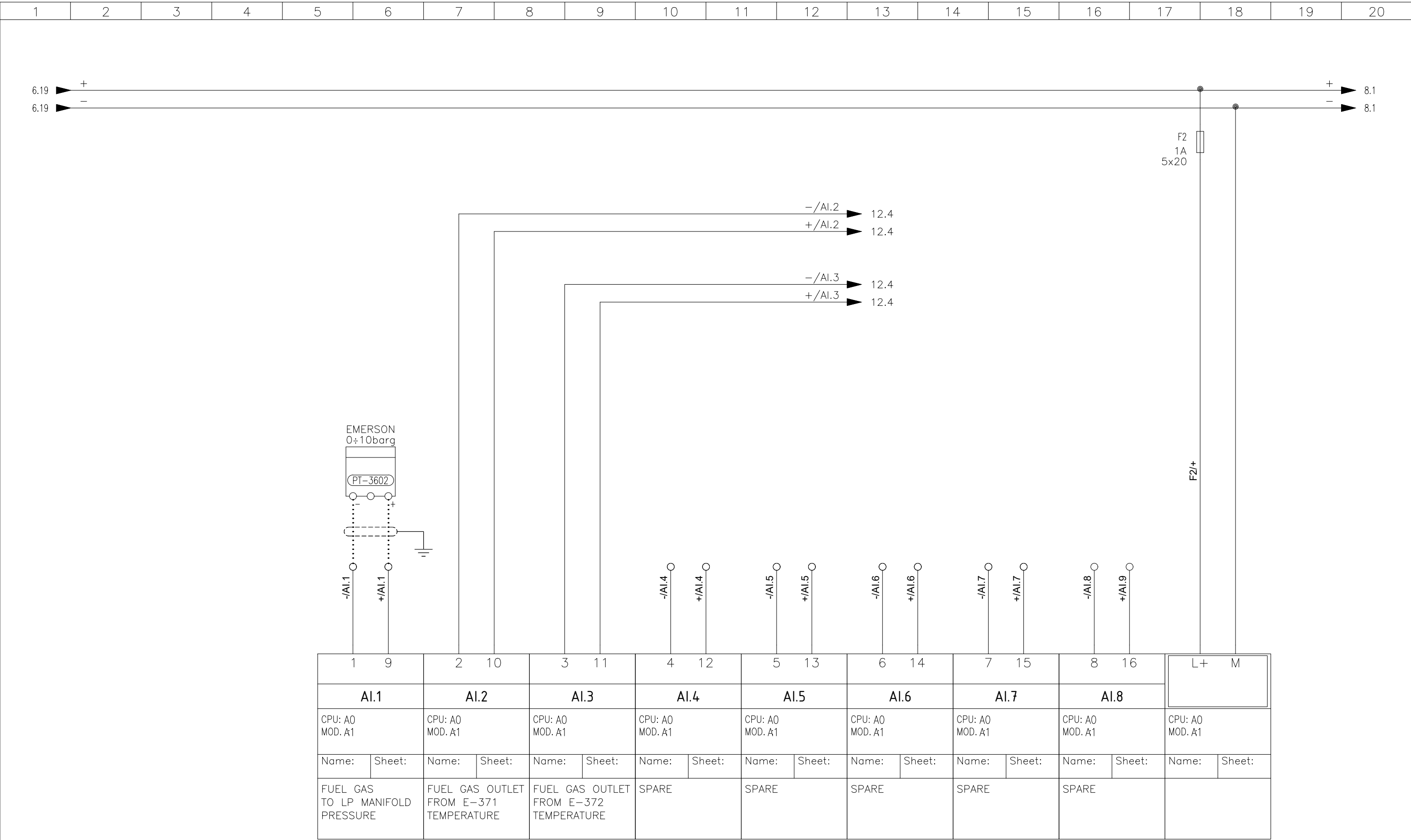
	POWER MAIN DISCONNECT SWITCH																	FEEDER ID	
	1F+N+PE~50Hz 230V																	RATED POWER P _n —RATED VOLTAGE V _n	
																		RATED CURRENT	

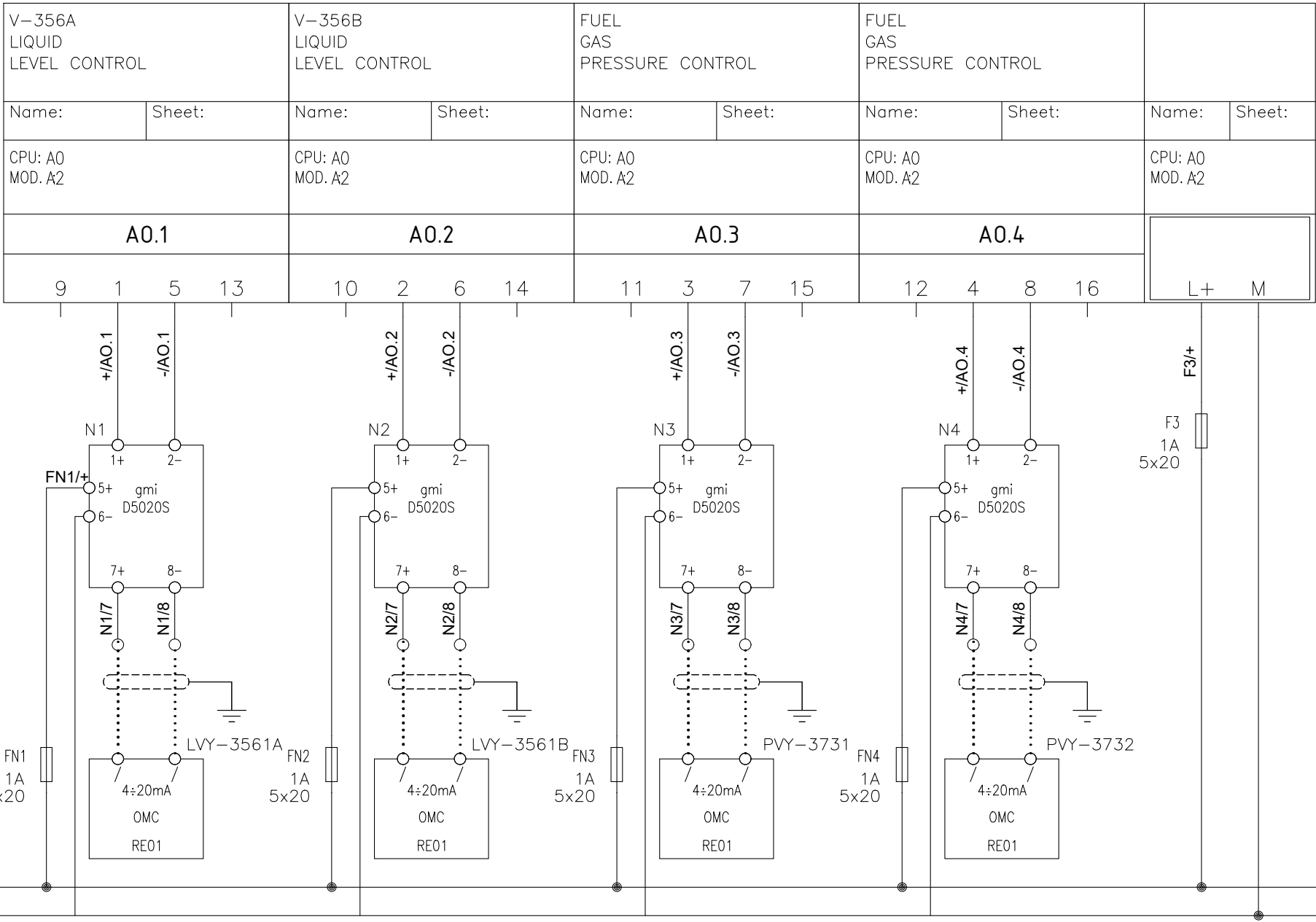
POWER SUPPLY FROM PCS

POWER INSTALLED: 240kW
I_n=10A I_B=1,1A
Fdp convenzional: 0,8
Factor Kc x Ku ≈ 1
POWER SUPPLY 1F+N+PE~50Hz 230V
I_{cc} Max=15kA
POWER ARMoured CABLE SECTION:
3Gx6mm²



CONDUCTOR PROTECTION	TECHNOR PSRC-220										TYPE	CONDUCTOR PROTECTION
	20A										RATED BREAKING CAPACITY Icu	
											RATED or THERMAL CURRENT In or Ith	
											MAGNETIC RELEASE CHARACTERISTICS Im	
											RATED R.C. SENSITIVITY Idn	
											WIRE SECTION	
											CABLE TYPE	
											CABLE CARRYING LOAD Iz	
											CABLE LENGHT	
											LINE ID	
										WIRE ID ON TERMINAL		





F3/+

F3

1A

5x20

7.19

+

7.19

-

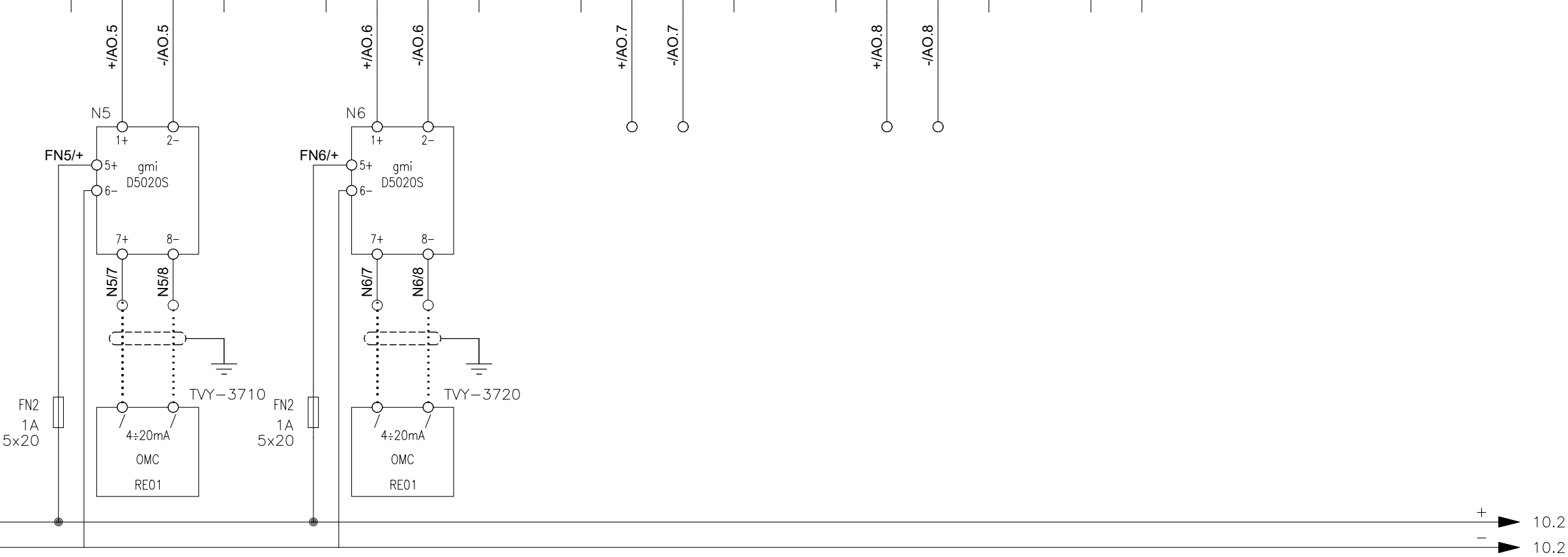
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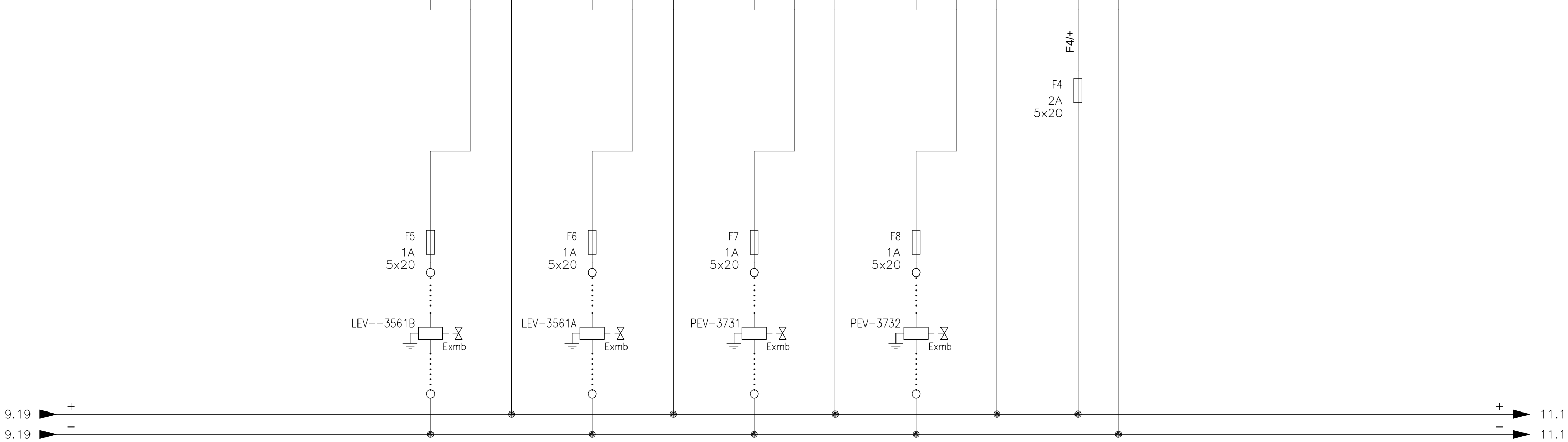
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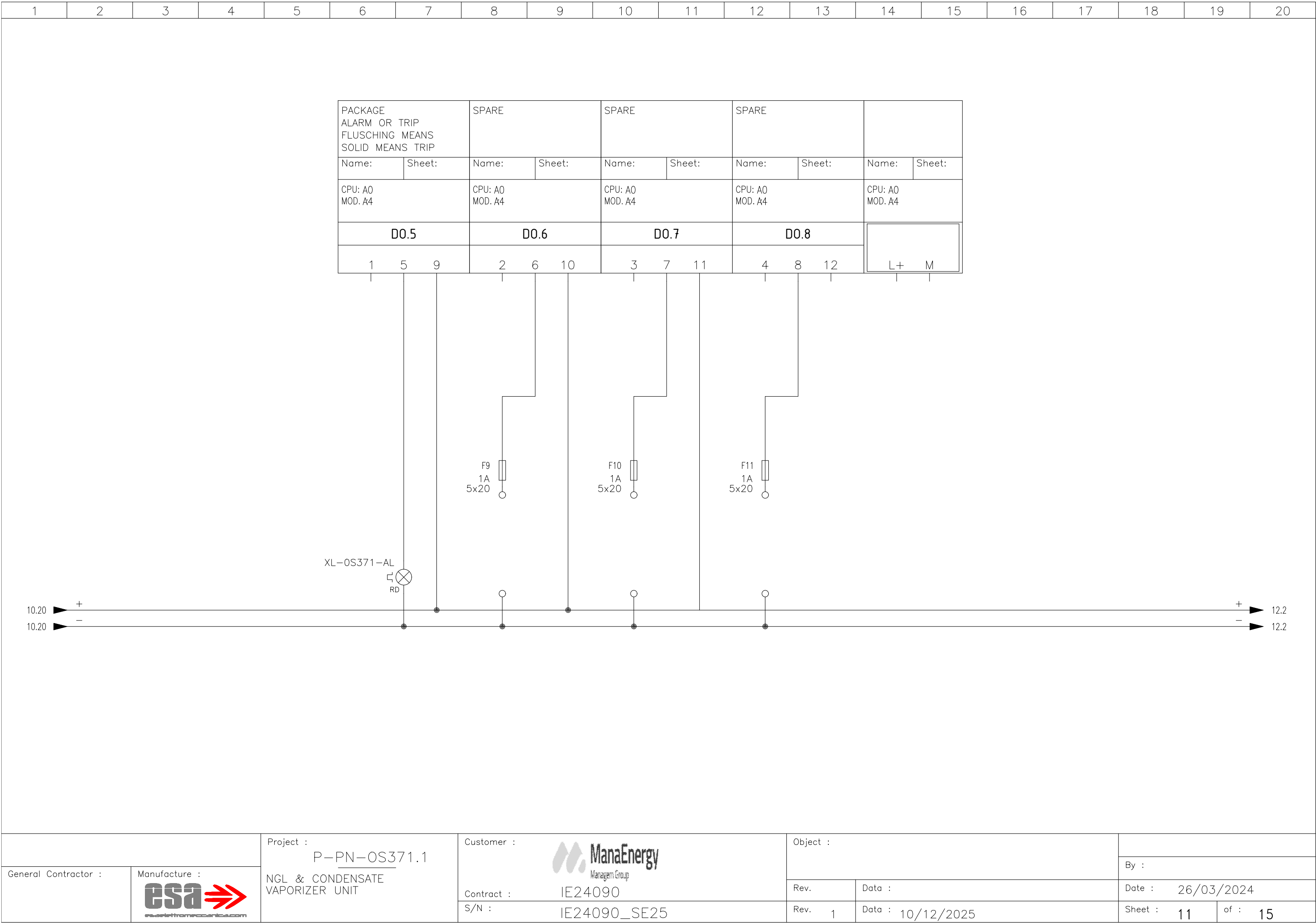
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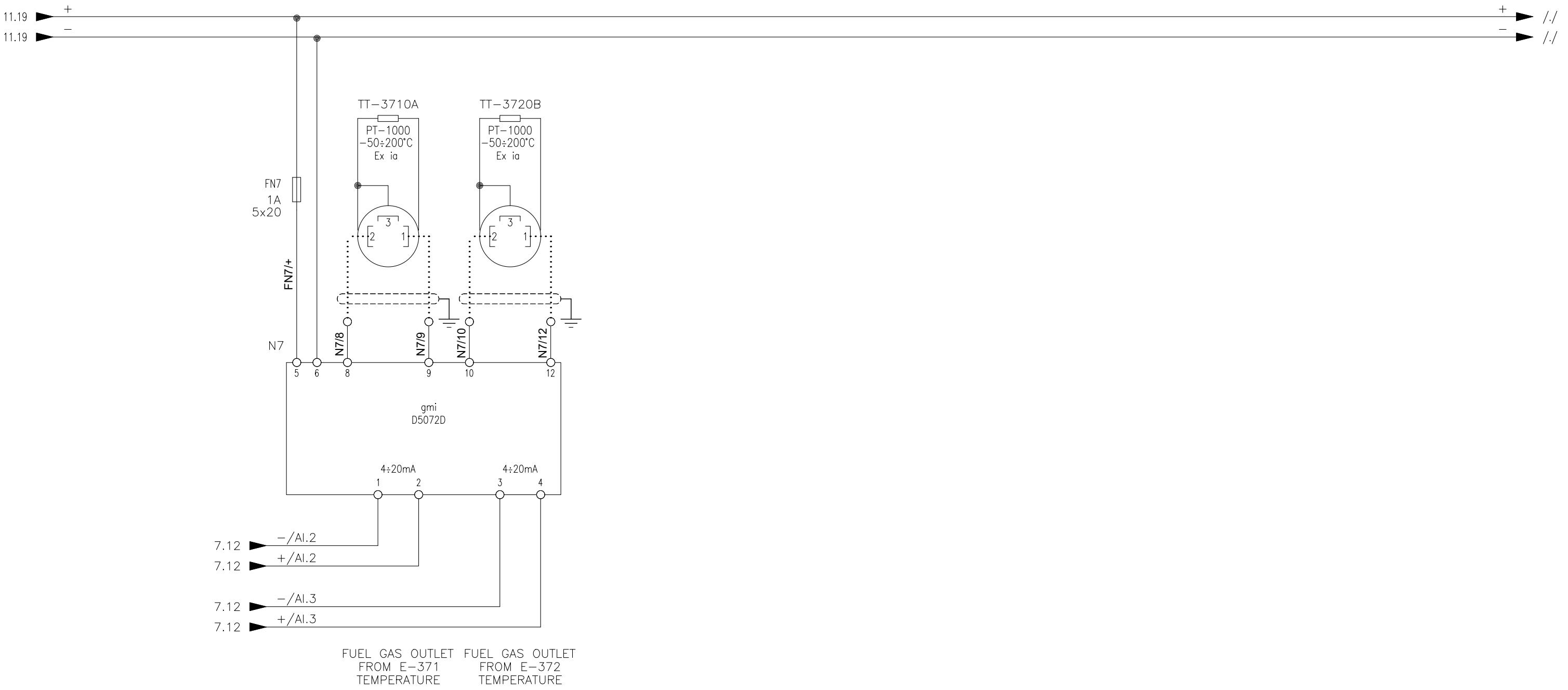
FUEL GAS OUTLET FROM E-371 TEMPERATURE CONTROL				FUEL GAS OUTLET FROM E-372 TEMPERATURE CONTROL				SPARE				SPARE							
Name:		Sheet:		Name:		Sheet:		Name:		Sheet:		Name:		Sheet:		Name:		Sheet:	
CPU: A0 MOD. A3				CPU: A0 MOD. A3				CPU: A0 MOD. A3				CPU: A0 MOD. A3				CPU: A0 MOD. A3			
A0.5				A0.6				A0.7				A0.8				L+M			
91513				102614				113715				124816							

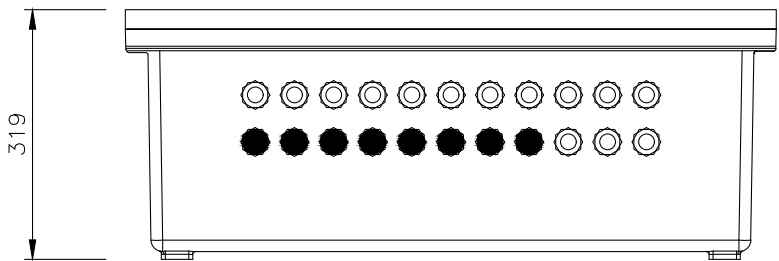
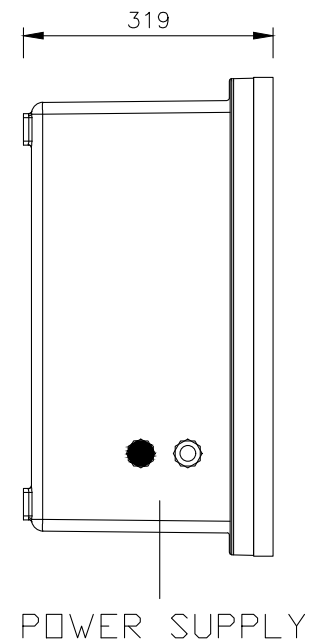
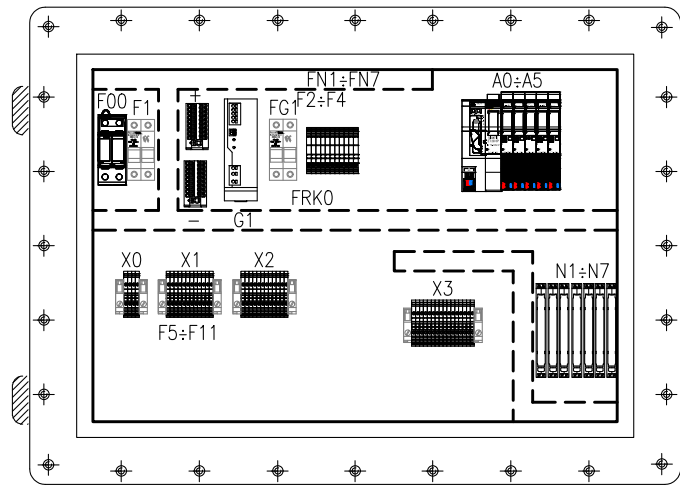
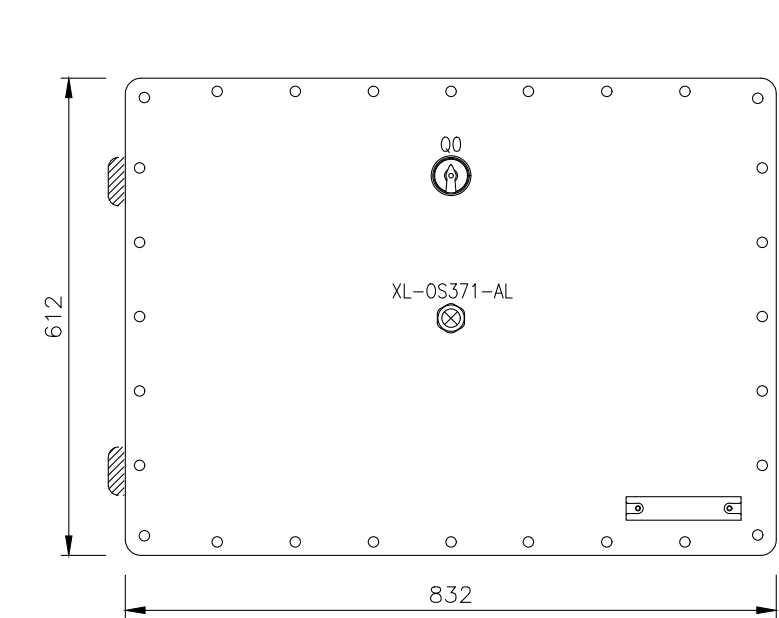


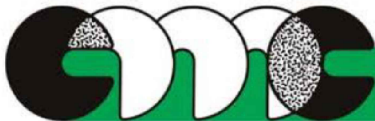
SHUT OFF VALVE LV-3561B			SHUT OFF VALVE LV-3561A			SHUT OFF VALVE PV-3731			SHUT OFF VALVE PV-3732					
Name:		Sheet:	Name:		Sheet:	Name:		Sheet:	Name:		Sheet:	Name:		Sheet:
CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5			CPU: A0 MOD. A5		
D0.1			D0.2			D0.3			D0.4			L+M		
159			2610			3711			4812					











C.M.C.
Antideflagranti s.r.l.

MILANO-ITALY Tel 024455358 Fax 0248401741

Type: EJB-G

230Vac 50Hz 16Å

Max Dis. P: 61 W

CE 0722

Ex II 2GD

INERIS18ATEX0026X/00

Ex db [ia Ga] IIB T6

Ex tb [ia Da] IIIC T85 °C IP66

-20°C < Tamb < +50°C

Year s.n 25/CMC/0173

Cable entries: see instructions

WARNINGS: DO NOT OPEN WHILE ENERGIZED

DO NOT OPEN IF AN EXPLOSIVE ATMOSPHERE MAY BE PRESENT

READ INSTRUCTIONS NOTE BEFORE INSTALLATION AND USE

INITIALS	SUPPLIER PN	DESCRIPTION	SUPPLIER
–	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
+	3273110	PTFIX 6/18X2,5–NS35 GY – Distribution block	PHOENIX CONTACT
A0	6AG1155–6AU01–7BN0	ET 200SP, PROFINET interface module IM 155–6PN Standard	SIEMENS
A0	6AG1193–6AR00–7AA0	ET 200SP BA 2xRJ45	SIEMENS
A2	6AG1134–6GF00–7AA1	AI 8xl 2–/4–wire BA	SIEMENS
A2	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A2	6AG1135–6HD00–7BA1	ET 200SP AQ 4xU/I	SIEMENS
A2	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A3	6AG1135–6HD00–7BA1	ET 200SP AQ 4xU/I	SIEMENS
A3	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals	SIEMENS
A4	6AG1132–6GD51–7BA0	ET 200SP RQ 4x24VDC/2A CO ST	SIEMENS
A4	6AG1193–6BP00–7DA0	ET 200SP BU15–P16+A0+2D push–in terminals	SIEMENS
A5	6AG1132–6GD51–7BA0	ET 200SP RQ 4x24VDC/2A CO ST	SIEMENS
A5	6AG1193–6BP00–7BA0	ET 200SP BU15–P16+A0+2B push–in terminals	SIEMENS
F00	952110	DG M TT 2P 275 – SPD	DEHN
F1	2302038	Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V	ITALWEBER
F10	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F11	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F4	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F5	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F6	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F7	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F8	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
F9	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FG1	2301138	Fuseholder for cylindrcal fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V	ITALWEBER
FN1	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN2	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN3	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN4	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN5	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN6	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FN7	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
FRK0	TBS4F	MODULAR TERMINAL WITH FUSE	CEMBRE
G1	2903149	TRIO–PS–2G/1AC/24DC/10 – Power supply unit	PHOENIX CONTACT
N1	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N2	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N3	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N4	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N5	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N6	D5020S	I.S. SIL2 HART? Isolating Driver	GMI
N7	D5072D	I.S. SIL2 Multifunction Temperature Converter	GMI
P–PN–OS371.1	EJB–H	ENCLOSURE 832x612x412 mm L x H x D	CMC
Q0	PSRC–320	SELECTOR SWITCH	CMC
XL–OS371–AL	PLD20	PILOT LAMP RED	CMC
XL–OS371–AL	O2OPD9	LAMPHOLDER BA9S, DIRECT SUPPLY	ELFI

